Voting Choice as a Predictor in County Level Sexual Orientation and Gender Motivated Hate Crimes across the United States: A Latent Growth Modeling Approach

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Preregistration and Data

Preregistration information can be found at https://osf.io/c9rw8, all other materials pertaining to this thesis can be found at https://github.com/LyraFunction/PSYC499.

Abstract

Animus towards LGBTQ (Lesbian, Gay, Bisexual, Transgender, and Queer) individuals remains high, exemplified through the growing incidence of hate crimes motivated by a victim's perceived LGBTQ identity. The increase in hate crime incidents may seem incongruous given recent polling data suggesting that attitudes towards the LGBTQ community are growing more positive year over year rather than negative. Recent work has proposed the importance of political factors influencing hate crime incidence in the United States, with jumps in occurrences loosely associated with major political events. Thus, my research sought to examine how one such event, the United States's 2016 presidential election, may relate to hate crime incidence for the following four years post-election. To answer my research question, I assessed whether hate crime incidence motivated by LGBTQ identities at a county level differed based on voting results in the 2016 election, such that Republican voting Republican counties had a higher incidence than Democratic counties over a four-year time span. Contrary to expectations, Republican counties had a lower average LGBTQ bias-motivated incidence compared to Democratic counties. Republican counties also had lower initial incidence and slower growth over time. While Republican counties had slower growth, the overall trend agnostic of county vote outcome was small but positive. When comparing individual bias motivations, average anti-gay incidence was significantly higher than all other bias motivations across all four years of analysis. These results suggest a more complicated relationship between election outcomes and bias-motivated incidence than previously implied, as there appear to be differences depending on what categories of bias are being analyzed.

Keywords: Hate Crime, LGTBQ, SOGI, Explicit Prejudice, Discrimination, Intergroup Aggression

Table of Contents

Intro....1
Population Counts of LGBTQ People....2
Explicit Prejudice....3
Reducing Prejudice....5
LGBTQ Experience of Hate Crimes....6
Politics as a Vehicle for Social Norm Change....9
The Current Research....11
Method....12
Election Data....12
Hate Crime Data....13
Data Preparation....14
Analytic Strategy....15
Figure 1....16
Results....16
Deviations from Preregistration....16
Missing Data....17
Descriptive Statistics....18
Table 1....18
Mean Comparisons....18
Figure 2....19
Latent Growth Model....20
Figure 3....20
Table 2....21
Exploratory Hypothesis....22
Figure 4....23
Figure 5....24
Discussion....25
Implications....28
Limitations....29
Conclusion....29
References....31
Appendix A....49
Table 3....49
Appendix B....51
Table 4....51
Appendix C....52
Table 5....52

Intro

The Supreme Court decision on *Obergefell v. Hodges* (2015), which required all states to license same-sex marriages while also recognizing any marriage licensed legally in other states. Following Obergefell, the GOP platform reaffirmed their opposition to same-sex marriage, explicitly laying out the party's intention to defend what they considered traditional marriage in their 2016 platform (*2016 Republican Party Platform*, 2016). This campaign promise created a rupture between the party and Log Cabin Republicans, one of the largest gay conversative organizations, with their president suggesting the 2016 platform was "the most anti-LGBT[1] platform in the party's 162-year history" (*Log Cabin Republicans on GOP Platform*, 2016).

While Donald Trump initially presented himself as a pro-LGBTQ candidate, especially in light of his opponents in the Republican primary leading up to the 2016 presidential election (Sosin, 2025). Shortly after being sworn in as president, federal agencies began removing mentions of the LGBTQ community (*Trump Administration Removes LGBTQ Content from Federal Websites*, 2017). Two months into Trump's first term, his administration revoked previous guidance protecting the rights of transgender students to use bathrooms that align with their gender identity (Johnson, 2017). Two years into the administration, Ben Carson, then secretary of the Department of Housing and Urban Development, defended the revocation of regulations that protected LGBTQ people from in housing and when attempting to access shelters for the unhoused (*Ben Carson Won't Restore LGBTQ Protections at HUD*, 2019). Ultimately, according to Cahill and Pettus (2020), the Trump administration enacted more anti-LGBTQ policies than any prior administration.

[1] This study will use LGBTQ (Lesbian, Gay, Bisexual, Transgender, and Queer), but when describing evidence from another source, I adopt the terminology used to avoid potentially overgeneralizing evidence. A more encompassing acronym that is more broadly used would be LGBTQIA+ (Lesbian, Gay, Bisexual, Transgender, Queer, Intersex, Asexual, with + referring to all gender/sexual identities not previously defined) ("Defining LGBTQ+," n.d.). I will use LGBTQ, as this acronym most accurately reflects the information provided by the datasets used in the current study.

Population Counts of LGBTQ People

Recent data from the 2020 United States census (US Census Bureau, 2020) reported 1,168,566 same-gender households. This number includes both married couples living together as well as unmarried, same-sex-attracted households. Of these reported households, nearly 75% (878,547) live within the 100 largest metropolitan areas (Spring & Ghaziani, 2024). Independent secondary analysis of the CDC's Behavioral Risk Factor Surveillance System conducted in 2023 suggested a much higher population of nearly 14 million with an average population of 5.5% across all states, including Washington, D.C. (Flores & Conron, 2023).

In comparison to the 2020 census, sampling completed by Gallup (2025) proposed a higher population, with 5.6% of 15,349 identifying as LGBTQ in 2020. Compared to 2016, this figure represented an increase of 1.4% (Gallup, 2025). This increase is largely driven by Gen Z (those born between 1997 and 2006) self-identification; since 2020, the percentage of Gen Z identifying as LGBTQ has increased from 18.8% to 23.1% (Gallup, 2025). On the other hand, older generations decreased in LGBTQ identification roughly by half within each generation, with 14.2% of Millennials identifying as LGBTQ, 5.1% of Generation X, 3% of Baby Boomers, and 1.8% of the Silent Generation (Gallup, 2025). Across all generations, a majority (56%) of the LGBTQ identifying individuals reported being bisexual, while 21% reported being gay, 15% reported being lesbian, 14% reported being transgender, and 6% of individuals reported something else (Gallup, 2025). While the general trend has increased, differences in population sizes exist on a state-by-state basis, with Vermont having the highest percentage of people identifying as LGTBQ at 5.3% in 2016 and South Dakota having the lowest at 2.0% (Gallup, 2017).

The gap between Gallup's sample and the Census data may be explained by the difference in the method of estimation: the census fails to count LGBTQ people living alone or otherwise not in the same household as their partner and overlooks bisexual people who tend to have cross-gendered partners (Pew Research Center, 2013; Wu et al., 2020). Difficulties and inconsistencies in the measurement of population sizes for the LGBTQ community may stem from three sources: the relatively small population of the community (Hughes et al., 2021), fear of disclosing an LGBTQ identity to researchers[2], and confusion among researchers regarding effective wordings on questionnaires for accurate identity assessment (Hughes et al., 2021). While LGBTQ people have been previously suggested to be difficult to survey by official sources (U.S. Census Bureau, 2016), limited evidence substantiates this claim (Bates et al., 2019). Suggesting that perceived barriers in collecting data from these populations may come from a lack of effort or from a lack of desire to include questions assessing LGBTQ characteristics (Cahill & Makadon, 2017).

Explicit Prejudice

LGBTQ identifying people frequently describe experiences of prejudice and discrimination (Meyer, 2003; Hurd & Mahal, 2025), with these experiences occurring as early as childhood (Russell et al., 2012). General support for LGBTQ+ rights has increased dramatically since early measurements, with 68% of polled Americans supporting gay marriage in 2020, compared to only 27% in 1996 (Gallup, 2025). While support for greater legal equality has increased, interpersonal attitudes have been less responsive and remained more negative. (Baunach 2011; Fetner 2016; Loftus 2001).

Often these types of experiences are based within an explicit attitude framework–an attitude that people know or are willing to report (Allport et al., 2015). This type of attitude can

[2] This may be especially true as it pertains to research conducted by governmental organizations (Pho et al., 2023).

be associated with three parts: a cognitive component based on assumptions about a group (i.e., stereotyping), an affective component relating to the emotions that come about when a prejudiced person interacts with another group, and a behavioral component toward the group (i.e., discrimination) (Jackson, 2020). Research on prejudice tends to use a more unidimensional definition of prejudice, operationalizing prejudice as simply a negative attitude toward one group of people (Hsieh et al., 2022; Paluck & Green, 2009). Newer scholarship has suggested that attitudinal approaches to sexual prejudice may capture two partially overlapping, but distinct, factors of moral disapproval and outgroup antipathy (Mallinas & Plant, 2024).

Between the incorporated groups of the LGBTQ community, differences exist between the relative ratings of each group, with bisexual individuals receiving worse ratings than lesbian and gay individuals where lower ratings indicate more negative attitudinal evaluations (Herek, 2002; Rye & Goldszmidt, 2025). When considering the ratings members of the LGBTQ community receive in comparison to one another, generally transgender individuals received lower ratings than bisexual individuals, who in turn received lower ratings than gay or lesbian individuals (Rye & Goldszmidt, 2025). Attitudes toward LGBTQ tend to vary based on various rater characteristics from age (Flores et al., 2020; Garretson, 2015), religious identity (Olson et al., 2006), emotional disgust (Gadarian & van der Vort, 2018; van Leeuwen et al., 2023; Wang et al., 2019), level of education (Budge, 2023; Flores, 2015), as well as gender (Flores et al., 2020; Herek, 2002; Kaufman & Compton, 2021; Rye & Goldszmidt, 2025). While religiosity (how often one attends religious services) may play an important role in explaining prejudicial attitudes targeting LGBTQ people (Rye & Goldszmidt, 2025), recent evidence suggests a more complex relationship depending on how accepting of LGBTQ people churchgoers perceive their church to be (Valen & Burke, 2025b).

Right-Wing Authoritarianism (RWA) and Social Dominance Orientation (SDO) are two constructs that are frequently examined in prejudicial research. These constructs are now more frequently identified as two dimensions of motivating factors that influence individuals' prejudice toward specific groups (Duckitt & Sibley, 2010). The first dimension, (RWA), measures one's submission to authority, adherence to traditional moral systems, and directed aggression towards outsiders (Altemeyer, 1981). The second dimension (SDO) is a measure of one's preference for a society that espouses hierarchy and inequality (Pratto et al., 1994), consisting of two distinct subdimensions (Ho et al., 2012). Each subdimension is highly correlated to one another but distinctly measures one aspect of the measure, with SDO-D measuring preference for group-based dominance and SDO-E assessing an aversion of equality (Ho et al., 2012). RWA and SDO have previously been thought of as personality traits due to their observed stability, but may more accurately be considered sets of stable social attitudes (Osborne et al., 2021). Both RWA and SDO share weak to strong associations with prejudicial attitudes towards LGBTQ individuals depending on the sample (e.g., Hatch et al., 2022; Tebbe & Moradi, 2012; Valen & Burke, 2025a; Whitley & Ægisdóttir, 2000).

Reducing Prejudice

Explicit prejudice tends to respond favorably to interventions intended to reduce it (Jayawickrama et al., 2025). Much of the previous literature on prejudice has examined how prejudice can be reduced, with significant attention paid to intergroup contact (e.g., DellaPosta, 2018; Hoffarth & Hodson, 2018). Previous meta-analytic evidence suggests a moderate but consistent reduction resulting from interactions with outgroups (Pettigrew & Tropp, 2006). These reductions through contact with an outgroup appear to result from enhancing knowledge of,

increasing empathy toward, and decreasing anxiety from contact with the outgroup (Panerati & Salvati; Pettigrew & Tropp, 2008).

Intergroup contact is not constrained to direct person-to-person interactions. While the effect is generally weaker compared to direct interactions, contact via media representations of target groups can effectuate prejudicial attitude reduction (Madžarević & Soto-Sanfiel, 2018). For example, contact with one member of the LGBTQ community may also function to reduce prejudicial attitudes towards other members through a positive secondary transfer effect, such as contact with a gay man reducing prejudice towards transgender men (Fitzgerald et al., 2024).

Intergroup contact is by no means a total prophylactic to prejudice, as all forms of contact are not equal, with the quality of contact appearing to have a stronger influence on reducing prejudicial attitudes than the quantity of contact events (Kanamori et al., 2022). Furthermore, the valence of contact plays a role, with negative interactions worsening attitudes and positive interactions bettering them (Affinito et al., 2023; Barlow et al., 2012). While reductional interventions for prejudice remain valuable, the malleable nature of prejudice also results in the possibility of prejudice increasing in response to certain events or experiences. One such way prejudice may be increased is by a negative secondary transfer effect, whereby prejudicial attitudes towards one outgroup generalize to another not immediately relevant outgroup (Affinito et al., 2023).

LGBTQ Experience of Hate Crimes

Despite renewed efforts to reduce bias, previous research demonstrate situations and contexts in which LGTBQ individuals face discrimination, such as acquiring housing (Glick et al., 2020), receiving health care (Puckett et al., 2018; Wall et al., 2023; Winter et al., 2016), and employment (Eames, 2024; James et al., 2016). LGBTQ people also face the possibility of hate

crimes, or criminal victimizations motivated by their identity, with experiences of hate crimes being relatively common among LGBTQ populations, especially for transgender individuals (Lombardi et al., 2002). Compared to the non-LGBTQ population, those identifying as LGBTQ are more likely to experience some form of victimization and are more than twice as likely for those victimizations to be bias-motivated (Flores et al., 2022). As with other forms of discriminatory experiences, those who experienced gender or sexual bias-motivated crimes were at higher risk to emotional harm and later physical harm (Holder, 2024).

Scholarship studying hate crimes has grown as data has become increasingly available and accessible, in large part due to governmental efforts to track crimes motivated by LGBTQ identity. Although an international definition of what constitutes a hate crime remains elusive (Vergani et al., 2024), research on hate crimes conducted in the United States commonly adopts the definition provided by Congress, which states that a hate crime is “a criminal offense against a person or property motivated in whole or in part by an offender’s bias against a race, religion, disability, ethnic origin, or sexual orientation” (Public Law 101-275, 1990). After expanding hate crime legislation to consider bias towards transgender individuals, transgender individuals reported increased reported support from families and the amount of people transgender people disclosed their identity to (Cramer et al., 2025).

While legislation has made large strides in researchers' ability to examine hate crime incidents in the United States, issues with underreporting remain pervasive (Giannasi & Hall, 2016). Indeed, analysis of the FBI’s general crime dataset from 2015 among all bias types suggested that victims were 6.2% less likely to report a crime if it was bias-motivated and 17.1% less likely to report incidence to police compared to non-bias-motivated crimes (Pezzella et al.,

2019). Underreporting may stem from fears that reporting hate crimes would result in further victimization (Haymes et al., 2000.; Peel, 1999).

Overall, hate crimes targeting LGBTQ people tended to increase in severity for increasing group minority status, such that LGBTQ women experience more severe hate crimes than their male counterparts, while non-white LGBTQ people experienced on average the most severe hate crimes (Coston, 2018). Additionally, recent evidence suggests that in comparison to religious, ethnic, or racially motivated hate crimes, crimes motivated by LGBTQ identities tend to be committed by people known to the victims[3], those under the influence of alcohol, or multiple perpetrators (Kehoe, 2020). Hate crimes motivated by LGBTQ identity are also more likely in rural areas as compared to urban areas (Piatkowska & Messner, 2022), concordant with general beliefs and reports regarding the acceptability of LGBTQ identification in rural areas (McKay et al., 2024). Research on the characteristics of hate crime offenders is limited, but among a sample of inmates in Spain those imprisoned for bias-motivated crimes were higher in generalized prejudice, homophobia and SDO than those imprisoned for non bias-motivated crimes (Pérez Ramírez et al., 2025).

The damage inflicted from suffering a hate crime tends to go beyond physical harm, with those victimized by bias-motivated crimes more likely to report experiencing psychological trauma (Fetzer & Pezzella, 2019). Among a sample of gay, bisexual, and other men who have sex with men, experiences of violence increased the likelihood of posttraumatic stress disorder features when participants attributed the experience to being motivated by their sexual identity; this was especially true for teenagers (Wiginton et al., 2023). For LGBTQ people who have

[3] Victim-perpetrator acquaintance may be a recent development, as older analysis of the relationship between LGBTQ victims of hate crimes and their perpetrators suggested that they were less likely to be victimized by those known to them rather than more (Herek, 2002). Another cause may be the expansion of coverage provided by hate crime legislation; to my knowledge, no scholarship has examined if the relationship between victims and perpetrators is associated with any particular gender or sexual identity.

experienced a hate crime personally, increases in anxiety and stress may be in part due to an internalization of homophobia (Burks et al., 2018). Furthermore, LGBTQ people who had direct experiences of hate crimes tended to have a decreased level of empathy towards others who experience hate crimes, as well as increased shame, anxiety, and feelings of vulnerability (Paterson et al., 2019).

Evidence on the motivations of hate crime perpetrators remains limited, but previous research by McDevitt et al. (2002) suggested that while bias-motivated crime perpetrators are motivated by an underlying bias against the target group, a majority of offenders were motivated by an additional desire to engage in thrill seeking via the act. A smaller proportion of perpetrators were motivated by their desire to "rid the world of groups they considered evil or inferior" or a desire to exact revenge upon the target group for a real or perceived crime inflicted upon the perpetrator's group (McDevitt et al., 2002).

Politics as a Vehicle for Social Norm Change

A noticeable jump occurred in reported hate crimes following the 2016 election (Edwards & Rushin, 2018), along with reporting suggesting that hate crime occurrence also increased in the specific counties in which Trump hosted his rallies (Feinberg et al., 2022). Crandall et al. (2018) found that compared to measures from before the election participants, regardless of whom they supported in the election, reported increased levels of prejudice towards groups targeted by Trump. Among Trump supporters, the observed increase was more pronounced compared to Clinton supporters, despite both groups having similar prejudice scores prior to the campaign (Crandall et al. 2018). Additionally, a prejudicial blind spot appeared, as while these participants scored higher on prejudice towards targeted groups, they perceived themselves to be less prejudiced than others (Crandall et al. 2018).

In response to the Supreme Court's decision in the Obergefell case, the perception of social norms shifted towards a more positive LGBTQ perception, without notable changes in individual attitudes (Tankard & Paluck, 2017). Likewise, research on the influence of state-level constitutional bans on same-gender marriage and hate crime incidence targeting LGBTQ individuals suggested a positive correlational relationship (Levy & Levy, 2017). Conversely, legal protections for same-gender marriage appear to be negatively associated with LGBTQ bias-motivated crime incidence (Nikolaou, 2022). Policies outlawing discrimination in the workplace on the basis of LGBTQ identities is also negatively associated with bias-motivated crime incidence. (Levy & Levy, 2017). Social norms may play an explanatory role in how individual attitudes pertaining to prejudice actualize, as among Trump supporters in a series of longitudinal studies prejudice towards groups targeted by the Republican candidate increased (Ruisch & Ferguson, 2022).

Compared to previous presidential elections, authoritarianism played a key role among voters in the 2016 election, with those high in the personality trait being over 50% more likely to vote for Donald Trump (Knuckey & Hassan, 2022). Furthermore, compared to those who opposed Donald Trump, those who favored him were significantly more likely to oppose gay marriage, legal protections for LGBTQ people, and support exceptions to anti-discriminatory policies for small businesses (Kaufman & Compton, 2021). Higher SDO and RWA relate to favorability and intention to vote for Trump election and lower intention to vote for Clinton in the 2016 (Choma & Hanoch, 2017). This relationship continued into the 2020 election, with SDO and RWA again predicting favorability and intention to vote for Trump (Godbole et al., 2022).

Authoritarianism has been presented as particularly important in understanding support for Trump, potentially eclipsing demographic characteristics (Macwilliams, 2016). High levels of RWA tend towards submission to authority figures. Thus, it is possible that those high in RWA, who simultaneously condemn hate crimes towards LGBTQ people, may experience an interpersonal shift in attitudes if they believe that political figures within their party have shifted to a less LGBTQ accommodating stance. Those high in SDO on the other hand, if they do not already hold a poor attitude toward LGBTQ individuals, may see a shift due to an increasing salience of the marginalization of LGBTQ identity exemplified through political rhetoric. Trump and the Republican party writ large may have been appealing to those high in authoritarianism due to repeated appeals to core triggers to authoritarianism such as the perception of outside threat (Feldman & Stenner, 1997). Perceptions of symbolic threat, which can be induced through manipulations of demographic size, in turn predict prejudicial attitudes toward LGBTQ people (Mackey & Rios, 2025). One manifestation of threat may be the perceived size of LGBTQ groups, which has some association with hate crime incidence motivated by LGBTQ identity (Piatkowska & Messner, 2022).

The Current Research

This study aimed to expand upon the current body of evidence by examining how a large socio-cultural event such as the 2016 election may have shifted social norms regarding the status of LGBTQ people. This shift may have fostered an environment where hate crimes began to appear more acceptable across the political spectrum, but, due to the differential voting patterns on a county-by-county basis, would have been experienced differently among county inhabitants. The “new” social norm would consequentially then manifest as a differential level of hate crimes motivated by LGBTQ identities. Over time this gap may widen over time as Trump and his

administration passed new policies that targeted LGBTQ rights, reaching its peak in the final year of his administration. To examine this theory I proposed the following three hypotheses:

H_1: Counties that voted for the republican candidate in the 2016 presidential election would have a higher average of hate crimes motivated by a LGBTQ identity compared to counties that voted for the democratic candidate.

H_2: The average amount of hate crimes motivated by LGBTQ identities across all counties in the United States would increase compared to the preceding timepoint.

H_3: County-level Republican vote majority in 2016 will positively predict both the intercept and slope of LGBTQ bias-motivated crimes, such that Republican-voting counties have higher initial levels of incidents and steeper growth over time than Democratic-voting counties.

I also tested one exploratory hypothesis:

E_1: Crimes motivated by anti-transgender and anti-gender non-conforming bias will be significantly higher than those motivated by anti-bisexual bias which will be significantly higher than those motivated by anti-lesbian or anti-gay bias at each timepoint

Method

Election Data

Data on election outcomes comes from the Massachusetts Institute of Technology Election Data and Science Lab (MEDSL) election returns dataset on United States presidential election results by county, using the data for the 2016 presidential election (MIT Election Data and Science Lab, 2018). The MEDSL dataset includes county-level vote totals for every state plus DC for the two major political parties, as well as an additional row for “other” candidates. The dataset also includes data on every presidential election from 2000-2024, but only the 2016 election data will be used. Of the total 3,158 counties included in the 2016 election, 3,155 will be

used for analysis. The three excluded counties are from Connecticut, Maine, and Rhode Island, which have a separate row counting votes from those in the military deployed internationally, or out-of-state voters.

Hate Crime Data

This study used the 2017-2020 hate crime incident data released by the FBI (United States Department of Justice. Federal Bureau of Investigation, 2022a, 2022b, 2023a, 2023b). The Uniform Crime Reporting Program (UCR) program defines hate crimes as any criminal offense that is at least partially motivated by the victim's race, religion, disability, sexual orientation, ethnicity, gender, or gender identity (United States Department of Justice. Federal Bureau of Investigation, 2022a). Importantly, even if the offender is mistaken about the victim's identity (i.e., the offender believes the victim to be bisexual despite them being straight), the crime is recorded as a hate crime motivated by the perceived identity of the victim (United States Department of Justice. Federal Bureau of Investigation, 2022a).

The FBI began collecting data on hate crimes following the passage of the 1990 Hate Crime Statistics Act, which required the agency to collect hate crime statistics with the ongoing UCR Program (Public Law 101-275, 1990). Since the agency began its hate crime reporting, several changes have occurred, increasing the types of bias covered, with bias against the disabled tracking beginning in 1997 and again in 2013 with the beginning of coverage for hate crimes motivated by gender, gender identity. In total, 27 motivations of bias are considered, broadly falling into the categories of race/ethnicity, religion, sexual orientation, disability status, gender, or gender identity.

While the UCR program has provided a great deal of information on the occurrence of hate crimes across the United States, agency participation in the program is voluntary, potentially

leaving gaps in coverage. Of the 18,623 policing-involved agencies (*Community Relations Service | 2020 FBI Hate Crimes Statistics*, 2021), only 16,149 participated in hate crime data collection in 2017 (*Hate Crime by Jurisdiction*, 2017). Agency participation fell 15,136 in 2020 (*Community Relations Service | 2020 FBI Hate Crimes Statistics*, 2021). However, following a change in the FBI's reporting system, population coverage fell to 70% for the duration of 2021 before recovering in 2022. Due to the incomplete coverage, some counties may be entirely missing data, possibly leading to a more limited sample size for data analysis. While agency participation fell, population coverage is consistently reported at around 95% from 2017 to 2020 (*2017 Hate Crime Statistics Released*, 2018).

Data Preparation

Due to the size and complexity of the datasets being used, I utilized code written in Python with the Pandas (The Pandas development team, 2025), NumPy (Harris et al., 2020), and Pyreadr (Fajardo, 2018) libraries for data manipulation, and file conversion. Using Python will ease the process of combining all the data used for this study. In total, nine separate data sources were merged into one larger dataset. Each dataset contains a FIPS code that denotes a specific county in the United States (*ANSI and FIPS Codes*, n.d.). Using these codes, I connected each dataset to one another. This combined dataset contained all the pertinent data for use in the data analysis. From the MEDSL dataset, I copied the county name, containing state, the party that won a given county (dummy-coded as 0-Democrat, 1-Republican), as well as the total votes each party received. The FBI UCR dataset contains two files; the first is a header file that contains every county that the FBI received information on, while the second file contains specific information on incidents. From the header file, I copied the county's population, if the county experienced any hate crime reports, which quarter any incidents took place in, and the incident

number if any exists. Finally, from the incident file, I copied the incident motivation(s) and the quarter in which the incident took place.

Analytic Strategy

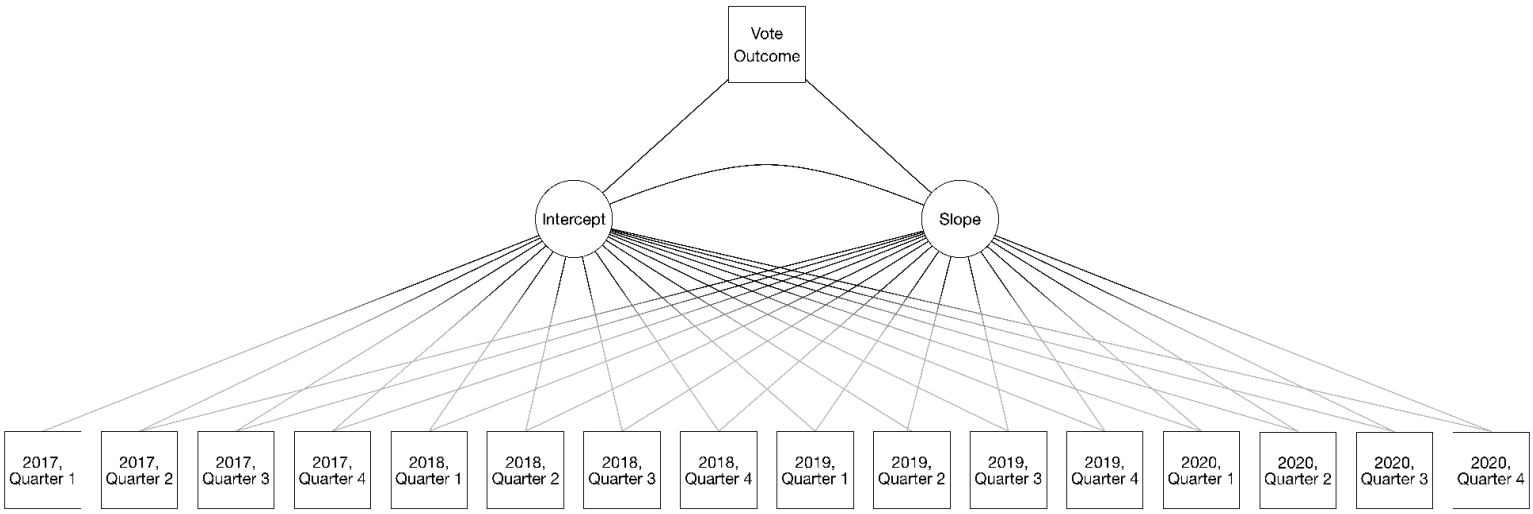
All analyses were performed in R (v4.4.2; R Core Team 2024). Once the datasets were combined I conducted descriptive analyses consisting of Mean, Standard deviation, Range, Skewness, and Kurtosis analyses for each year by election outcomes. To test hypothesis one and two, I initially planned to use a mixed-effects ANOVA, but due to assumptions issues expanded upon later, I instead opted to use a series of Bonferroni adjusted Wilcox tests for group comparisons and Mann-Whitney for year to year comparisons.

In order to test hypothesis three, I constructed a Latent Growth Model with the lavaan package in R (Rosseel et al., 2012). LGM is an implementation of Structural Equational Modeling (SEM) that handles the analysis of longitudinal data, capturing the growth trajectory of a variable over a set period of time. Using a LGM allowed for the examination of potential changes in hate crime growth over a set period on the results of county-level voting. My conceptual model (see Figure 1) consists of one categorical predictor variable, the voting results for a given county in the 2016 election, where 0 signifies a democratic majority and 1 indicates a republican majority, two latent variables—the slope and intercept for changes in hate crimes—and 16 indicators consisting of the four quarters of hate crime reports targeting LGBTQ individuals across the four years of interest (2017, 2018, 2019, 2020).

A LGM measures intra- and inter-county change in hate crime incidence, allowing for a robust analysis of any potential change in observed incidence. Although there is a significant degree of mathematical resemblance between mixed-effect ANOVAs and LGMs (McNeish & Matta, 2018), the addition of a LGM to a mixed-effects ANOVA enables a more robust

management of measurement error through the modeling of latent variables (Office for National Statistics (ONS), 2022). By modeling the latent intercept and slope, I can account for situations where growth in hate crime incidence (or lack thereof) is due to greater or lesser starting incidence (Duncan & Duncan, 2009). Additionally, LGMs have greater flexibility regarding data with a pronounced percentage of zeros (Seddig, 2024).

Figure 1.
Conceptual diagram of Latent Growth Model



Results

Deviations from Preregistration

Due to unexpected issues regarding the shape of the data deviations from preregistration had to be made. While I initialled planned to perform a mixed-effects ANOVA with t-tests as a backup, the data failed the meet the homogeneity of variances (Levene's Test, $p < 0.001$) as well as the sphericity assumption for both the party between-subjects variable ($p < .001$), and year within-subjects variable ($p < .001$). Additionally, the data were substantially positively skewed and leptokurtic (see Table 1). These factors suggested that a mixed-effects ANOVA would be inappropriate for use in analyzing the dataset. Instead I chose to use a Wilcox test for group

comparisons for each year, and a Mann-Whitney test for year to year comparisons. Regarding my exploratory hypothesis, I had originally planned to only conduct a comparison between LGBTQ bias categories using each category's average value over the four years; I was concerned that examining the data for each year individually would result in excessive zeros. When preparing for the analysis the averaged data already contained excessive zeros meaning the data was unsuitable for analysis with a one-way ANOVA. I instead elected to run the analysis for all each year of data included in the providing information on how the LGBTQ bias categories compare to one another at each timepoint instead of collating the info using a conover test.

Initially, I planned to exclude chi-square fit indices for the Latent Growth Model as chi-square tests suffer bias from large sample sizes (Bollen, 1989, 1990; Finch & French, 2015), but the degrees of freedom can provide some diagnostic information on models beyond the chi-square value, thus I elected to include the test to provide additional information on the model. Lastly, while my latent growth model used dummy-coding to delineate democratic and republican voting counties, I also ran the model again with effects coding applied (Republican voting counties coded as 1, Democratic counties coded as -1). The results for the effect coding change had little overall change and identical fit by BIC, CFI, TLI, and RMSEA, but not through the chi-square test ($\chi^2(136) = 78340.903$ $p < .001$, RMSEA = .157 [.155 -.159] $p > .05$, CFI = .858, TLI = .866, BIC = 60171.462). Full results for the alternative coding are viewable in Table 5 within the appendix.

Missing Data

Data was considered missing when: there was a mismatch between MEDSL dataset FIPS codes and UCR FIPS codes, the MEDSL dataset listed vote totals for a county as zero, or the UCR had entirely missing data for a county. Under this criteria several counties in Alaska; DC;

Kansas City, Missouri; Bedford, Virginia; and the boroughs of New York City were removed. See Appendix A for a complete list. Ultimately, of the maximum of 3,155 counties, 3,109 counties were valid and retained for data analysis.

Descriptive Statistics

Once missing data were addressed, I calculated the mean, standard deviation, range, skewness, and kurtosis statistics (see Table 1) for the total bias-motivated incidence of each year by county voting. Additionally, I calculated the mean, and standard deviations for each bias-motivated category (anti-gay, lesbian, bisexual, transgender, gender-nonconforming, and general LGTBQ bias) displayed in Table 1. I also calculated a correlation matrix for each bias category per year which can be found under Appendix B, Table 4.

Mean Comparisons

As discussed previously, instead of the planned mixed-effects ANOVA to test hypotheses one and two, I opted instead to use a series of Bonferroni corrected Wilcox and Mann-Whitney tests. After correcting for the multiple comparisons, the alpha value for the following analyses was set to .007. To determine if Republican voting counties reported higher average LGBTQ bias-motivated incidents compared to Democratic counties, I ran four Wilcoxon signed-rank

Table 1.

Descriptive Statistics of LGBTQ Bias-Motivated Incidents by Year and County Winner

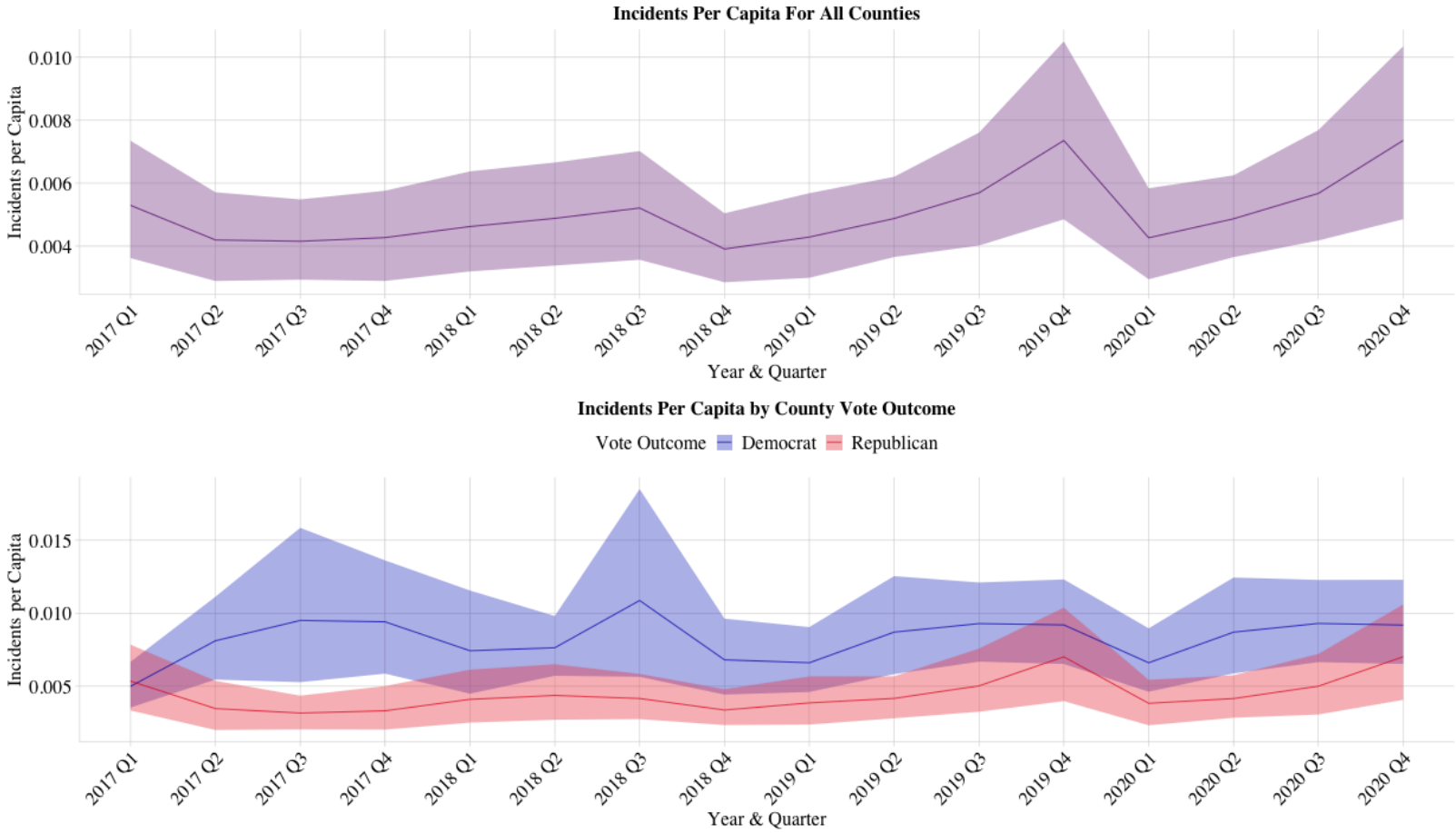
Party		*M*	*SD*	*Range*	*Skew*	*Kurtosis*
2017	Democrat	.34	.97	0 – 11.06	5.64	43.27
	Republican	.17	1.04	0 – 20.65	11.02	155.37
2018	Democrat	.35	1.07	0 – 15.49	8.04	91.1
	Republican	.17	.91	0 – 17.89	9.32	116.92
2019	Democrat	.38	1.12	0 – 17.32	8.84	113.87
	Republican	.20	1.21	0– 30.42	12.41	220.26
2020	Democrat	.38	.98	0 – 11.36	5.92	50.02
	Republican	.25	1.23	0 – 19.57	8.79	98.94

Note: Mean values should be interpreted in terms of incidents per 10,000 people. For example, in 2017, on average counties that voted for the Hilary Clinton in the 2016 election reported .34 LGBTQ bias-motivated incidents per 10,000 people.

tests, one for each year of analysis. Each year of analysis had small effect sizes and significant differences, but with a direction opposite to what was anticipated: 2017, W = 8438841, $p < .001$ $r = .279$ [.23 – .33]; 2018, W = 8413969, $p < .001$, $r = .292$ [.24 – .34]; 2019, W = 8343291, $p < .001$, $r = .277$, [.23 – .32]; 2020, W = 8230788, $p < .001$, $r = .257$, [.21 – .30].

Figure 2.

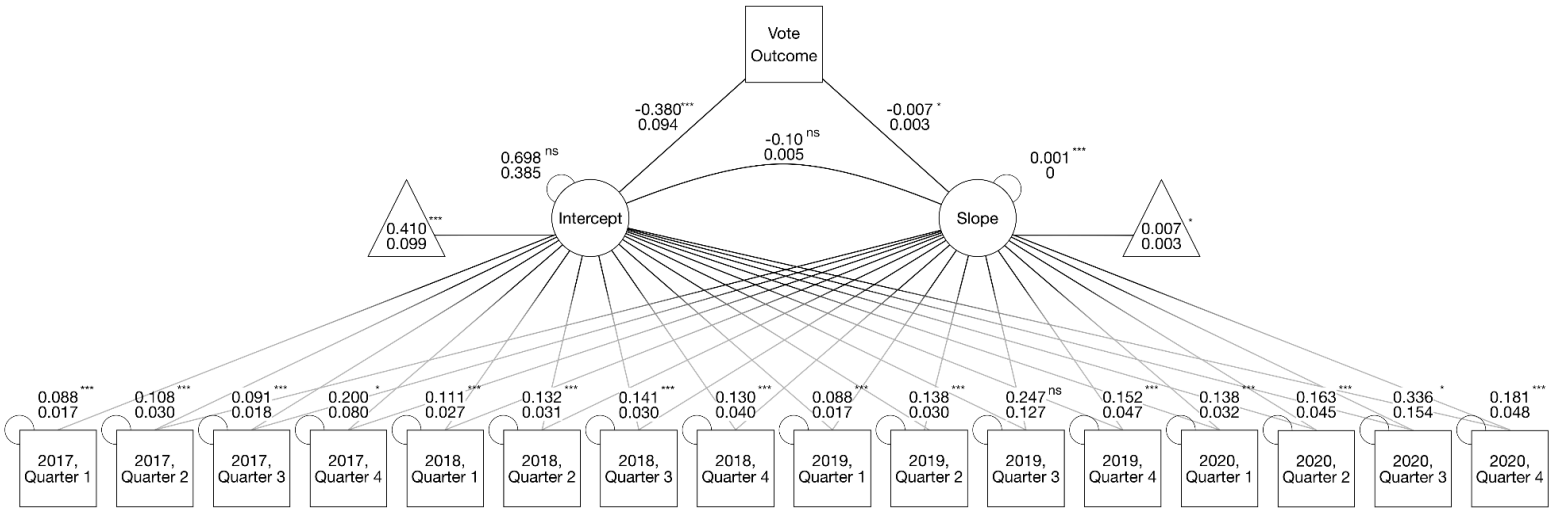
Means of Bias-Motivated Incidents By Quarter



Note: The darker lines represent mean values while the lighter ribbon represents a 95% confidence interval.

In regards to hypothesis two, I ran a series of paired Mann-Whitney tests to determine if the average LGBTQ bias-motivated incidents was higher than the previous year. There were no significant differences in incidence between 2017 and 2018 (V = 62416, $p = 0.2342$ r = .0144, [.00081 – 0.05]), 2018 and 2019 (V = 71821, $p = .159$, r = .0104 [.00054 – 0.05]), or 2019 to

Figure 3.
Path Diagram for Latent Growth Model.



Note: Results are unstandardized, for standardized results see Table 3.

2020 (V = 78926, p = .01712, r = .0354 [.0044 – 0.07]). While the p value for the 2019 to 2020 comparison is lower than the traditional alpha of .05, the value is lower than the corrected alpha for the current study and is consequently not significant.

Latent Growth Model

To test hypothesis three, I used a Latent Growth Model with the Lavaan package in R. The model fit the data after 70 iterations of the Maximum Likelihood estimator. The model exhibited poor fit of data by chi-square and RMSEA fit indices ($\chi^2(144) = 11235.216$ $p < .001$, RMSEA = .157 [.155 – .160] $p > .05$), but acceptable fit by CFI and TLI fit indices (CFI = .858, TLI = .866, BIC = 60171.462). A full visualization of the model output can be seen in Figure 4.

At the initial timepoint, 2017 Quarter 1, the intercept (B = 0.410 [0.234 – 0.630], SE B = 0.099, β = .485, $p < .001$), and slope were significant (B = 0.007 [0.001 – 0.014], SE B = 0.003, β = .303, p = .022) indicating that LGBTQ bias-motivated incidents were increasing overtime,

with a moderate effect size. The latent slope and intercept did not significantly covary (*B* = -0.010 [-0.023 – -0.001], SE B = 0.005, β = -0.492, *p* = .067), suggesting that a counties initial value had little effect on the overall slope of change in bias-motivated incidence over time. The variance of the latent intercept was not significant (*B* = 0.698 [.973 – .973], SE B = 0.385, β = .973, *p* = .07), suggesting that counties across the United States were generally similar in their reported incidents at the initial timepoint. Conversely, the slope variance was significant (*B* = 0.001 [0.000 – 0.001], SE B = 0, β = .990, *p* = .001), indicating that counties had large deviances in slope trajectories. Residual variances for individual timepoints, as well as latent standardized values, can be seen in Table 3 below, but generally, regarding the residuals, many of the time points have significant unexplained variance.

A county voting for Trump in the 2016 election was associated with lower initial latent intercept (*B* = -0.380 [-0.582 – -0.210], SE B = 0.094, β = -.164, *p* < .001) and slower growth regarding the latent slope (*B* = -0.007 [-0.014 – 0], SE B = 0.003, β = -.098, *p* = .047). These results suggest republican voting counties were more likely to have lower initial values and slower growth over time, compared to democratic voting counties.

Table 2.

Latent Growth Model Results

Effect		*Estimate*	*SE*	*95% CI*		*Std.lv*	*Std.All*	*p*
		B	SE B	*LL*	*UL*		β	
Regressions								
Intercept ~ Party		-0.380	0.094	-0.582	-0.210	-.449	-.164	< .001
Slope ~ Party		-0.007	0.003	-0.014	0	-.269	-.098	.047
Intercepts								
Intercept		0.410	0.099	0.234	0.630	.485	.485	< .001
Slope		0.007	0.003	0.001	0.014	.303	.303	.022
Covariance								
Intercept ~ Slope		-0.010	0.005	-0.023	-0.001	-.492	-.492	.067
Variances								
2017	Quarter 1	0.088	0.017	0.056	0.123	.088	.150	< .001
	Quarter 2	0.108	0.029	0.058	0.170	.108	.134	< .001

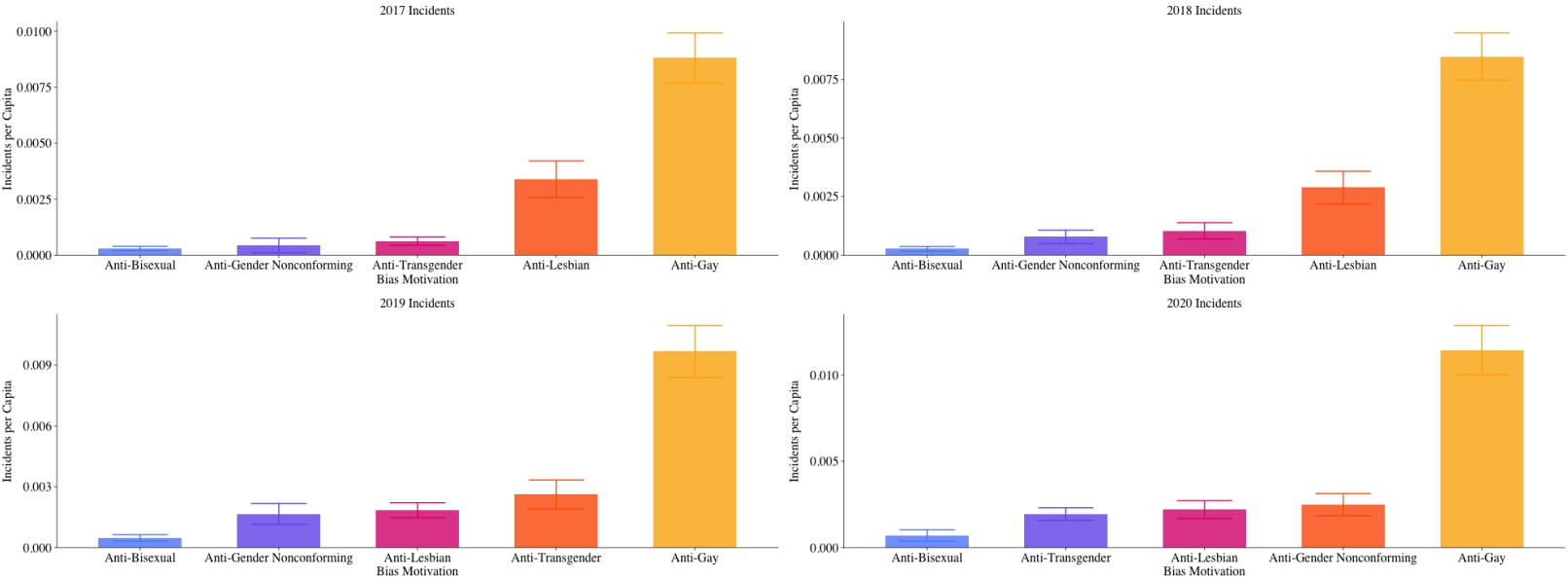
	Quarter 3	0.091	0.018	0.061	0.131	.091	.118	< .001
	Quarter 4	0.200	0.077	0.078	0.388	.200	.231	.01
2018	Quarter 1	0.111	0.026	0.069	0.168	.111	.146	< .001
	Quarter 2	0.132	0.032	0.077	0.207	.132	.172	< .001
	Quarter 3	0.141	0.028	0.092	0.205	.141	.185	< .001
	Quarter 4	0.130	0.038	0.072	0.218	.130	.175	.001
2019	Quarter 1	0.088	0.017	0.057	0.124	.088	.128	< .001
	Quarter 2	0.138	0.029	0.084	0.205	.138	.190	< .001
	Quarter 3	0.247	0.123	0.088	0.527	.247	.298	.046
	Quarter 4	0.152	0.045	0.078	0.254	.152	.209	.001
2020	Quarter 1	0.138	0.029	0.086	0.198	.138	.194	< .001
	Quarter 2	0.163	0.042	0.089	0.247	.163	.223	< .001
	Quarter 3	0.336	0.149	0.106	0.676	.336	.374	.024
	Quarter 4	0.181	0.045	0.103	0.275	.181	.244	< .001
Intercept		0.698	0.385	0.111	1.591	.973	.973	.07
Slope		0.001	0	0	.001	.990	.990	.001

Exploratory Hypothesis

A Kruskal-Wallis test indicated significant differences between hate crime incidents among LGBTQ groups in 2017 (H(4) = 403.416, $p < .001$), further results from a post-hoc Conover test revealed significant differences between anti-bisexual incidents and anti-gay incidents ($t = -16.706$, $p < .001$), anti-bisexual incidents and anti-lesbian incidents ($t = -5.433$, $p < .001$), anti-gay incidents and anti-gender nonconforming incidents ($t = 17.609$, $p < .001$), anti-gay incidents and anti-lesbian incidents ($t = 11.272$, $p < .001$), anti-gay incidents and anti-transgender incidents ($t = 14.993$, $p < .001$), anti-gender nonconforming incidents and anti-lesbian incidents ($t = -6.337$, $p < .001$), and anti-lesbian incidents and anti-transgender incidents ($t = 3.722$, $p = .001$). No significant differences were found between anti-bisexual incidents and anti-transgender incidents ($t = -1.712$, $p = .435$), anti-gender nonconforming incidents and anti-transgender incidents ($t = -2.615$, $p = .045$), anti-bisexual incidents and anti-gender nonconforming incidents ($t = 0.903$, $p > .05$).

A Kruskal-Wallis test indicated significant differences between hate crime incidents among LGBTQ groups in 2018 ($H(4) = 383.142$, $p < .001$), further results from a post-hoc

Figure 4.
Differential Incident Rates of LGBTQ Bias-Motivations by Year.



Note: Bars represent 95% confidence intervals.

conover test revealed significant differences between anti-gay incidents and anti-transgender incidents (t = 14.357, p < .001), anti-gay incidents and anti-gender nonconforming (t = 16.693, p < .001), anti-gay incidents and anti-lesbian incidents (t = 11.343, p < .001), anti-bisexual incidents and anti-gay incidents (t = -16.944, p < .001), anti-bisexual incidents and anti-lesbian incidents (t = -5.600, p < .001), anti-gender nonconforming and anti-lesbian incidents (t = -5.350, p < .001). No significant differences were found between anti-bisexual incidents and anti-gender nonconforming (t = -0.251, p > .05), anti-bisexual incidents and anti-transgender incidents (t = -2.587, p = .046), anti-gender nonconforming and anti-transgender incidents (t = -2.336, p = .096), and anti-lesbian incidents and anti-transgender incidents (t = 3.0135, p = .013).

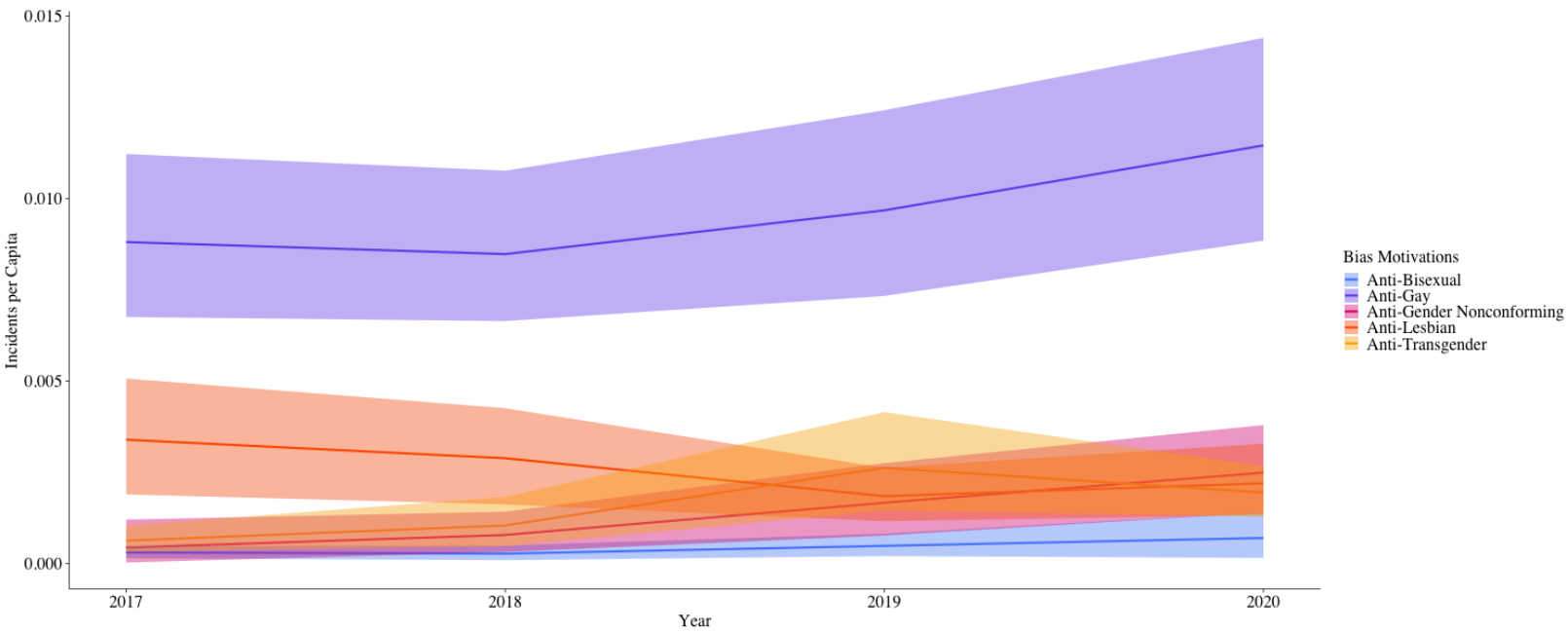
A Kruskal-Wallis test indicated significant differences between hate crime incidents among LGBTQ groups in 2019 (H(4) = 322.080, p < .001), further results from a post-hoc conover test revealed significant differences between anti-gay incidents and anti-gender nonconforming (t = 14.628, p < .001), anti-gay incidents and anti-lesbian incidents (t = 12.118, p

< .001), anti-gay incidents and anti-transgender incidents (t = 12.796, p < .001), anti-bisexual incidents and anti-gay incidents (t = -16.084, p < .001), anti-bisexual incidents and anti-lesbian incidents (t = -3.966, p < .001). No significant differences were found between anti-bisexual incidents and anti-gender nonconforming (t = -1.456, p = .727), anti-bisexual incidents and anti-transgender incidents (t = -3.288, p > .05), anti-gender nonconforming and anti-lesbian incidents (t = -2.510, p = .060), anti-gender nonconforming and anti-transgender incidents (t = -1.8319, p = .335), anti-lesbian incidents and anti-transgender incidents (t = 0.678, p > .05).

A Kruskal-Wallis test indicated significant differences between hate crime incidents among LGBTQ groups in 2020 ($H(4)$ = 383.142, p < .001). Further analysis from a post-hoc conover test revealed significant differences between anti-gay incidents and anti-gender nonconforming incidents (t = 15.060, p < .001), anti-gay incidents and anti-lesbian incidents (t = 12.762, p < .001), anti-gay incidents and anti-transgender incidents (t = 12.275, p < .001), anti-bisexual incidents and anti-gay incidents (t = -17.499, p < .001), anti-bisexual incidents and anti-lesbian incidents (t = -4.7358, p < .001), and anti-bisexual incidents and anti-transgender incidents (t = -5.222, p < .001). No significant differences were found between anti-bisexual incidents and anti-gender nonconforming incidents (t = -2.438, p = .074), anti-gender nonconforming incidents and anti-lesbian incidents (t = -2.298, p = .108), anti-gender nonconforming incidents and anti-transgender incidents (t = -2.783, p = .026), and anti-lesbian incidents and anti-transgender incidents (t = -0.486, p > .05).

Figure 5.

LGBTQ Bias-Motivations Over Time Year.



Discussion

The purpose of this study was to model the relationship between the 2016 election and the reporting of LGBTQ bias-motivation incidents in the subsequent four years once Trump entered the White House. Contrary to my three primary hypotheses, Republican voting counties reported significantly lower average LGBTQ bias-motivated incidents compared to their Democrat voting county counterparts. Additionally, regarding the Latent Growth Model, Republican voting counties had lower initial values and a lower slope than Democrat voting counties. The exploratory hypothesis also provided unexpected results, I anticipated that bias-motivated incident reports would be roughly congruent with warmth ratings for LGBTQ identities — wherein transgender people are perceived less warmly than bisexual people, who are in turn perceived colder than gay and lesbian people — instead across all four years analyzed, gay bias-motivated incidents were consistently the highest.

Given the unexpected results, I believe there are several possible explanations worth considering and that could be tested with additional data. One plausible explanation may be that

differences in incidence are a result of differences in LGBTQ population distribution between Republican and Democratic voting counties. That is to say, more LGBTQ individuals reside in states and counties that lean democratic (Spring & Ghaziani, 2024). Therefore the number of incidents against LGBTQ individuals would be higher as a result of opportunity and proximity. Alternatively, if the true LGBTQ bias incidence rate in the United States differs from the pattern observed in this study, a plausible explanation could be that underreporting disproportionately affects Republican counties. Undercounting can result from multiple sources, from a distrust in police to reporting agency error (Pezzella et al., 2019), if Republican counties are indeed underreporting bias motivated incidence there may be a gap between Democratic and Republican counties on reporting officer knowledge. On the other hand, a mix of these explanations could exist wherein some counties are — either purposefully or accidentally — underreporting bias-motivated incidence while others simply do not have a population of LGBTQ people large enough for even those motivated to perpetuate the acts.

Previous bias-motivated incidence work has demonstrated the association of relative population sizes of racial groups relative to the majority group (Cikara et al., 2022). LGBTQ populations tend to cluster in specific states, and more specifically in select cities (Spring & Ghaziani, 2024), meaning that rural counties that tend to vote Republican may simply lack both the opportunity — due to small LGBTQ population sizes — and the possible motivation, if the majority group lack a sufficient feeling of threat by LGBTQ populations. If population size provided the best explanation for county level differences, given the currently available data, the results of the exploratory hypothesis are especially unexpected, as bisexual people constitute the majority of those who identify as LGBTQ (Gallup, 2025), yet bisexual bias-motivated incidence

was significantly lower than gay bias-motivated incidence which may suggest that population size alone is not a sufficient explanation for the observed pattern of bias-motivated incidence.

Another consideration for LGBTQ bias-motivated incidents, even when assuming positively motivated reporting officers, is that victim identities may present challenges for accurate categorization of offenses. Consider the following scenario from the Hate Crime Data Collection Training manual (2022) released by the FBI to assist local reporting agencies:

> A Transgender woman was walking down the street near her home when three men walking toward her said, “Hey what’s your problem? Huh?” She kept walking, trying to ignore them. However, as they got close, one yelled “We don’t want no queers in this neighborhood!” and a second one knocked her to the ground. (p. 18).

The training manual suggests this scenario to be classified as a simple assault motivated by anti-transgender bias. While the manual cites an outdated definition of queer as an anti-gay slur (Worthen, 2023), its suggestion is correct in the motivation for the offense. Yet, the plausibility of this scenario is questionable, as its categorization relies on the reporting officer to have known the identity of the offense's target. LGBTQ identities tend to be perceived as highly concealable (Le Forestier et al., 2025), which potentially limits the accuracy of outside observers reports if they are not the direct target of a bias-motivated offense, creating downstream effects on reports. Additionally, given that reporting of incidents is in part predicated on evidence found at the scene of a crime such as witness statements or graffiti of slurs used, further issues may arise for accurate categorization of offenses as LGBTQ groups may differ when it comes to knowledge of slurs targeting incorporated identities (i.e., knowledge of slurs for bisexual individuals may be less available than slurs targeting lesbian women).

Underreporting may also result due to the issues illustrated previously, as a result of reporting agency bias, or merely the perception of bias. Nearly a third of victims of bias-motivated crimes report an absence of confidence in police agencies, and more than a fifth report that they chose to report their experience to another official (Pezzella et al., 2019; see also Bell, 2002; Lantz and Wenger, 2022). These statistics are unsurprising, as those targeted by bias-motivated crime are demographically similar to those who report high levels of police-related stress, potentially due to historical or current feelings of being over-policed or targeted by police for their identity (Bacchus et al., 2023). The issue of agency bias is correctable, as recent evidence suggests that the presence of bias units, specialized divisions within a reporting agency trained on understanding bias-motivated incidents, increases reporting of such incidents (Piatkowska et al., 2026). While these units may increase reporting of public incidents, they may have limited impact in increasing reports of incidents that are less public, such as incidents where victims know the perpetrator a scenario LGBTQ targets of bias-motivated incidents are at an increased probability to experience (Herek, 2002).

Implications

Hate crimes represent a severe physically (Fredriksen-Goldsen et al., 2014; Miller & Denise, 2015; Saghir et al., 1970) and physiological (Almeida et al., 2009; Amadio, 2006; Bailey, 1999; Fredriksen-Goldsen et al., 2014) detrimental behavioral outcome motivated by a person's marginalized identity, and understanding what motivates a perpetrators actions could have powerful impacts on the lives of those at risk. This study contradicts the previous literature concerning the relationship between election outcomes and bias-motivated crime incidence by examining one grouping of motivations. The bulk of the literature instead focuses on collating incidence, which, considering the observed results, may point to differential relationships

between motivations.

Limitations

The current study has several limitations that should be kept in mind when interpreting the results. Primarily, this study used results on a county level, removing the ability to interpret how individuals may have acted or felt following the 2016 election, and those changes may or may not have influenced their behavior regarding gender sexual minorities. Instead, the results only allow for a broad understanding of how the election may be associated with reported bias-motivated incidence. Additionally, while temporal precedence was established in this study, due to the lack of individual-level measurement, it would be inappropriate to attempt to draw casual conclusions from the data reported. As such, future research may wish to measure attitudes on an individual basis and measure associations with aggressive behavior tasks such as the Taylor Aggression Paradigm (Taylor, 1967) with manipulations for the other "player".

Another promising avenue for the future concerning bias-motivated incidence may be to include additional parameters. These parameters may include additional election information, such as results from midterm or local elections, or variables not considered in the current study, such as economical information about a county. Unfortunately, when it comes to using the FBI's hate crime statistics for future research on bias-motivated incidence, researchers may be limited in their options as the population coverage of the dataset has been trending downwards since January 2025. As of March 2026, coverage dropped from 95% in December 2024 to 19%. This means that of those agencies required to report, only 19% are currently in compliance. This will result in skewed analyses since there will be missing data. .

Conclusion

I examined whether LGBTQ bias-motivated incidence differed over the course of the

sixteen quarters starting in 2017 and the end of 2020 based on a county's voting outcome in the 2016 election. Though I did not find evidence to support my hypotheses, instead finding the opposite pattern than expected. I believe that future research examining additional election outcomes or comparing discrete bias categories to one another may help shed additional light on the impact of current rhetoric within the political sphere. Additionally, I urge and hope that agencies will continue to provide data for future analyses to be conducted.

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Appendix A.

Table 3.

Counties Unsuitable for Analysis

State	County Name	FIPS Code	Reason for Removal
N/A	District of Columbia	11001	Invalid population count (0)
New York	Bronx	36005	Invalid population count (0)
New York	Kings	36047	Invalid population count (0)
New York	New York	36061	Invalid population count (0)
New York	Queens	36081	Invalid population count (0)
New York	Richmond	36085	Invalid population count (0)
Virginia	Bedford	51515	Invalid vote total and population count (0)
Missouri	Kansas City	2938000	Invalid provided FIPS code
Alaska	District 1	2001	MEDSL provided FIPS code does not match UCR code
Alaska	District 2	2002	Code mismatch
Alaska	District 3	2003	Code mismatch
Alaska	District 4	2004	Code mismatch
Alaska	District 5	2005	Code mismatch
Alaska	District 6	2006	Code mismatch
Alaska	District 7	2007	Code mismatch
Alaska	District 8	2008	Code mismatch
Alaska	District 9	2009	Code mismatch
Alaska	District 10	2010	Code mismatch
Alaska	District 11	2011	Code mismatch
Alaska	District 12	2012	Code mismatch
Alaska	District 13	2013	Code mismatch
Alaska	District 14	2014	Code mismatch
Alaska	District 15	2015	Code mismatch
Alaska	District 16	2016	Code mismatch
Alaska	District 17	2017	Code mismatch
Alaska	District 18	2018	Code mismatch
Alaska	District 19	2019	Code mismatch
Alaska	District 20	2020	Code mismatch
Alaska	District 21	2021	Code mismatch
Alaska	District 22	2022	Code mismatch
Alaska	District 23	2023	Code mismatch
Alaska	District 24	2024	Code mismatch
Alaska	District 25	2025	Code mismatch
Alaska	District 26	2026	Code mismatch
Alaska	District 27	2027	Code mismatch
Alaska	District 28	2028	Code mismatch

Alaska	District 29	2029	Code mismatch
Alaska	District 30	2030	Code mismatch
Alaska	District 31	2031	Code mismatch
Alaska	District 32	2032	Code mismatch
Alaska	District 33	2033	Code mismatch
Alaska	District 34	2034	Code mismatch
Alaska	District 35	2035	Code mismatch
Alaska	District 36	2036	Code mismatch
Alaska	District 37	2037	Code mismatch
Alaska	District 38	2038	Code mismatch
Alaska	District 39	2039	Code mismatch
Alaska	District 40	2040	Code mismatch
Alaska	District 99	2099	Code mismatch

Note. The exclusion of many of the counties in Alaska may be due to the MEDSL election dataset using newer FIPS code definitions than the UCR dataset. Since the MEDSL election dataset includes data on presidential elections up to the 2024 election, they may update FIPS codes for older elections to modern equivalents for comparisons.

Appendix B.

Table 4.

Descriptive Statistics and Correlations for Bias-Motivated Incidents by Year

Motivation	*M*	*SD*	*1*	*2*	*3*	*4*	*5*	*6*
2017								
Anti-Lesbian	.04	.29	1					
Anti-Gay	.20	2.06	.71***	1				
Anti-Bisexual	.01	.10	.15***	.22***	1			
Anti-Transgender	.03	.45	.52***	.73***	.24***	1		
Anti-Gender nonconforming	0	.07	.31***	.33***	.13***	.43***	1	
Anti-General	.1	1.04	.31***	.40***	.22***	.62***	.61***	1
2018								
Anti-Lesbian	.04	.35	1					
Anti-Gay	.22	2.06	.75***	1				
Anti-Bisexual	.01	.09	.08***	.08***	1			
Anti-Transgender	.03	.40	.55***	.71***	.01	1		
Anti-Gender nonconforming	.01	.11	.25***	.25***	-.01	.36***	1	
Anti-General	.10	1.43	.34***	.34***	.06***	.59***	.50***	1
2019								
Anti-Lesbian	.04	.41	1					
Anti-Gay	.24	2.21	.83***	1				
Anti-Bisexual	.01	.11	.09***	.12***	1			
Anti-Transgender	.04	.40	.69***	.79***	.03	1		
Anti-Gender nonconforming	.02	.23	.25***	.31***	.24***	.30***	1	
Anti-General	.10	.71	.51***	.66***	.16***	.58***	.35***	1
2020								
Anti-Lesbian	.04	.32	1					
Anti-Gay	.25	2.02	.52***	1				
Anti-Bisexual	.01	.11	.17***	.08***	1			
Anti-Transgender	.06	.78	.35***	.84***	.07***	1		
Anti-Gender nonconforming	.02	.21	.16***	.34***	.05**	.34***	1	
Anti-General	.12	.77	.36***	.43***	.14***	.41***	.29***	1

Note: All statistics are rounded to the nearest two decimal places, leading to some values showing as zero.

* $p < .05$ ** $p < .01$ *** $p < .001$

Appendix C.

Table 5.

Latent Growth Model Results with Effects Coding

Effect	*Estimate*	*SE*	*Std.lv*	*Std.All*	*95% CI*		*p*
					LL	*UL*	
Regressions							
Intercept ~ Party	-0.190	0.048	-.225	-.164	-0.291	-0.102	< .001
Slope ~ Party	-0.003	0.002	-.134	-.098	-0.007	-0.000	.043
Intercepts							
Intercept	0.220	0.048	.260	.260	0.131	0.325	< .001
Slope	0.004	0.002	.169	.169	0.001	0.007	.013
Covariance							
Intercept ~ Slope	-0.010	0.005	-.492	-.492	-0.023	-0.001	.067

Note. Variances are equivalent to those in the dummy coded model thus not included here.